Drought Risk Analysis

1. Introduction & Data Overview Climate change is no longer a future projection but a present reality for Turkey. Positioned in the Mediterranean basin, the country faces unique vulnerabilities. This Exploratory Data Analysis (EDA) aims to quantify the environmental shifts and their correlation with agricultural output.

The Dataset The analysis integrates three primary data streams:

Thermal Data (T2M): Average temperature at 2 meters above the earth's surface (°C).

Hydrological Data (PRECTOTCORR): Corrected precipitation amounts (mm/day).

Agricultural Value: The monetary value of crop production (Thousand TL) at the provincial level, aggregated into regional totals.

The data was wrangled to ensure temporal consistency and spatial alignment, transforming raw provincial Excel files and regional CSVs into a "Tidy Data" format for robust visualization.

2. Hydrological Shifts: Precipitation Volatility While temperature shows a steady climb, precipitation is characterized by its unpredictability. For an agrarian economy, the distribution of water is as critical as the total amount.

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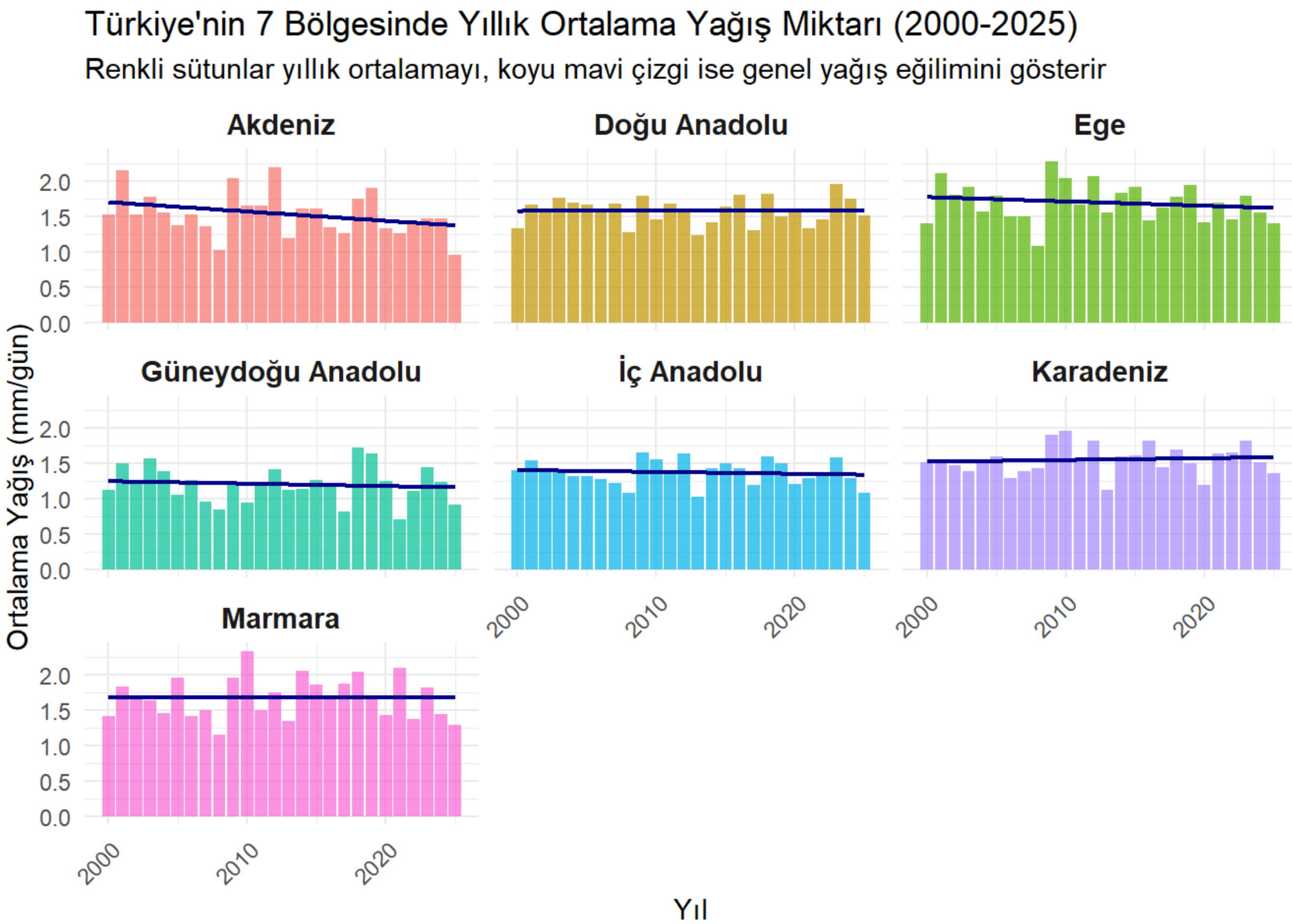

Analysis of Water Stress The Black Sea region remains the outlier with high and relatively stable precipitation. However, the Aegean and Central Anatolia regions—Turkey's primary grain and fruit baskets—show concerning levels of volatility. The downward or stagnant trend lines in these regions, combined with rising temperatures, indicate a high risk of "evapotranspiration," where plants lose moisture faster than the soil can replenish it.

Next stage of our EDA explores how the economic value of crops has changed alongside these climatic shifts.

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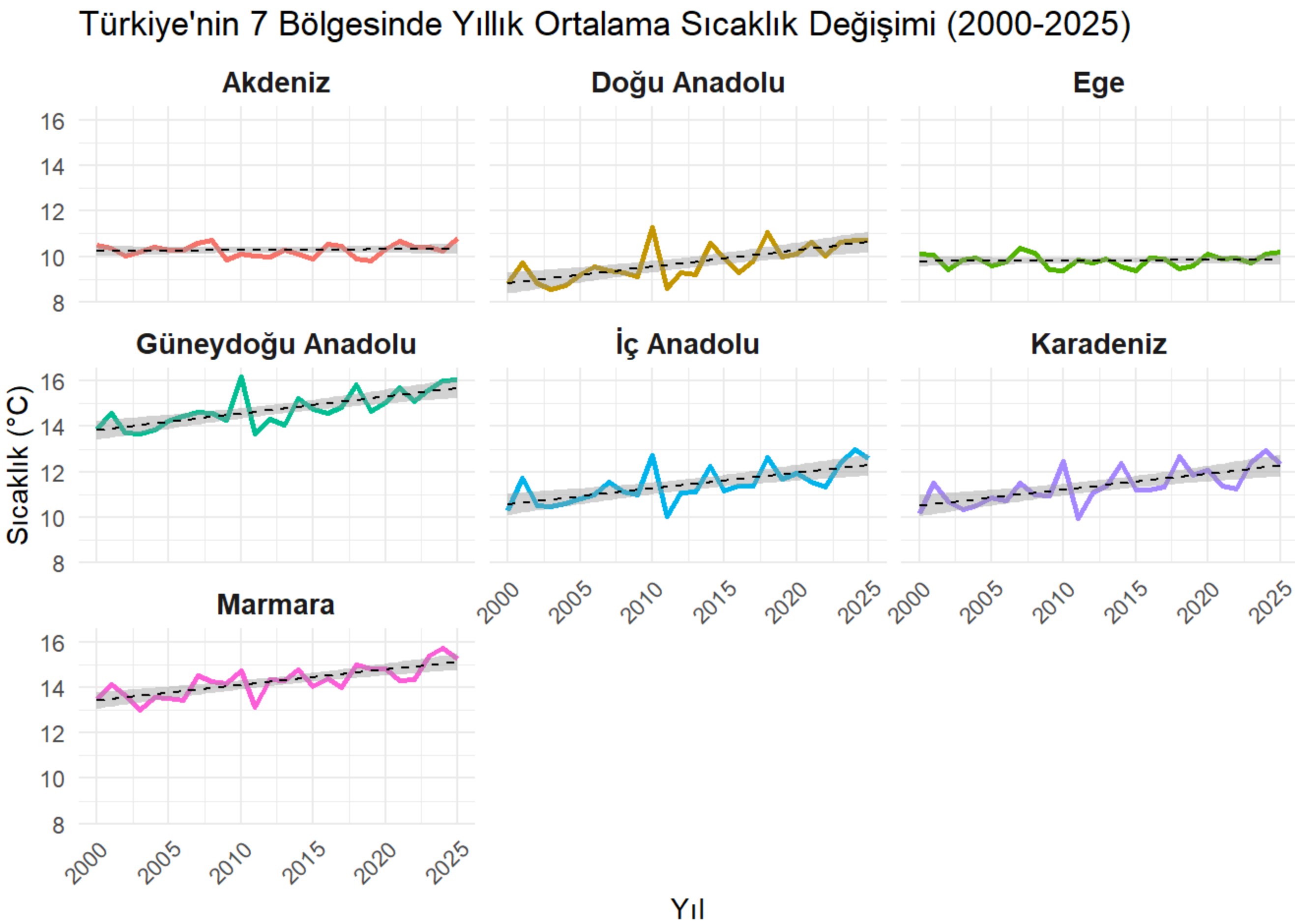

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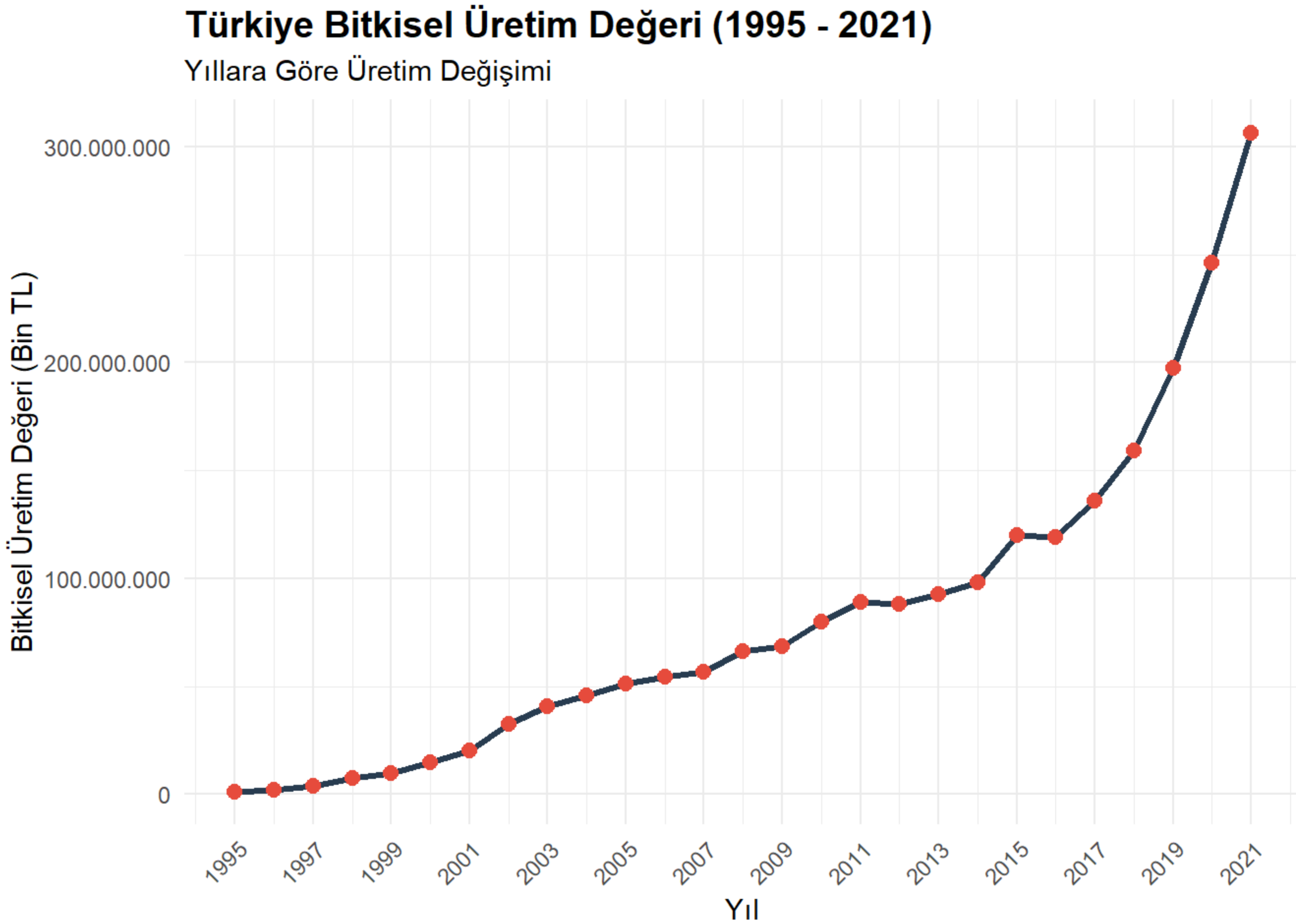

The nominal value of agricultural production has grown exponentially, especially in the Mediterranean and Aegean regions. While this appears positive, it must be viewed with caution. This rise is likely influenced by inflation and the shift toward high-value, export-oriented greenhouse crops. The divergence between “rising value” and “deteriorating climate conditions” suggests that Turkish agriculture is becoming increasingly capital-intensive, requiring more investment in irrigation and fertilizers to combat climate stress.

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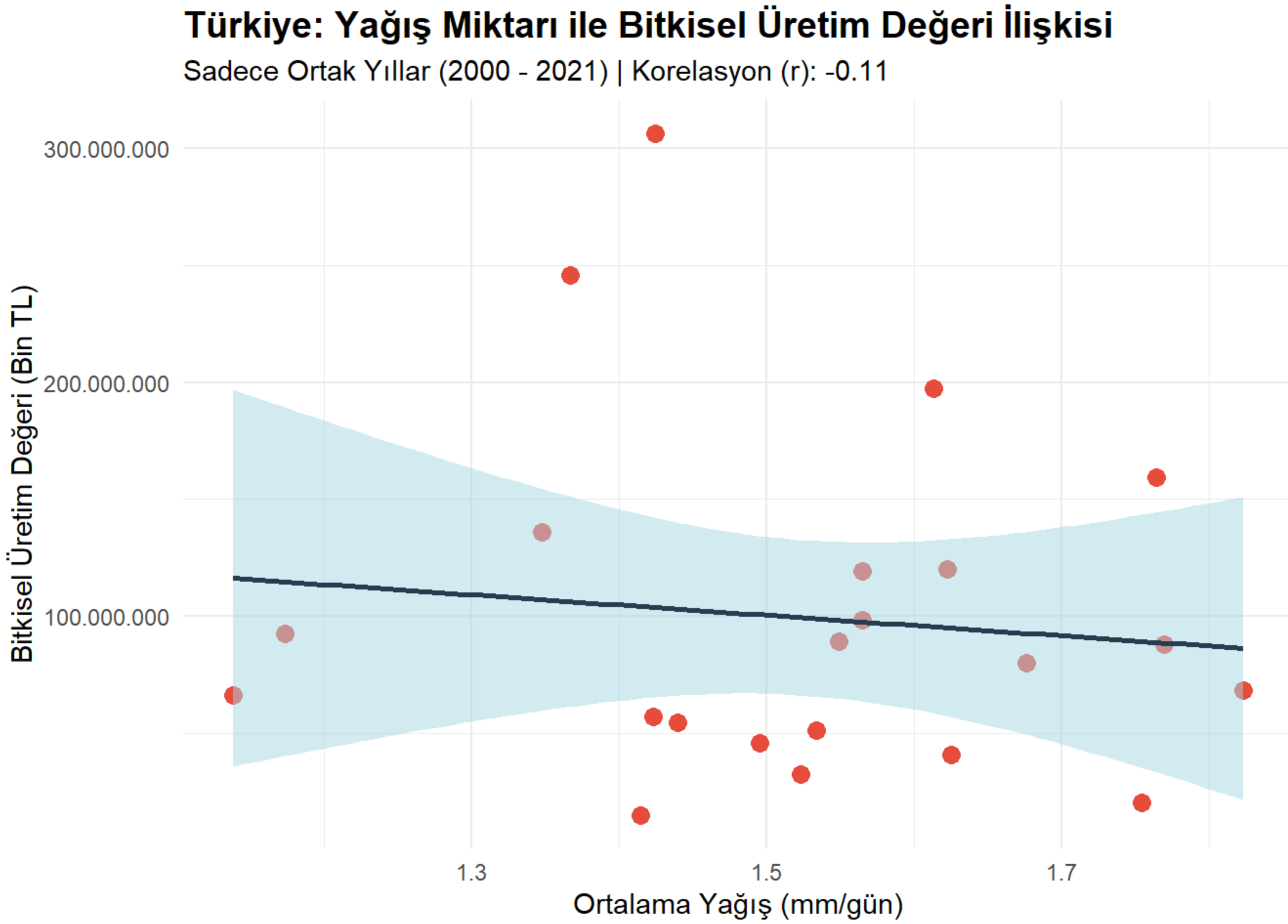

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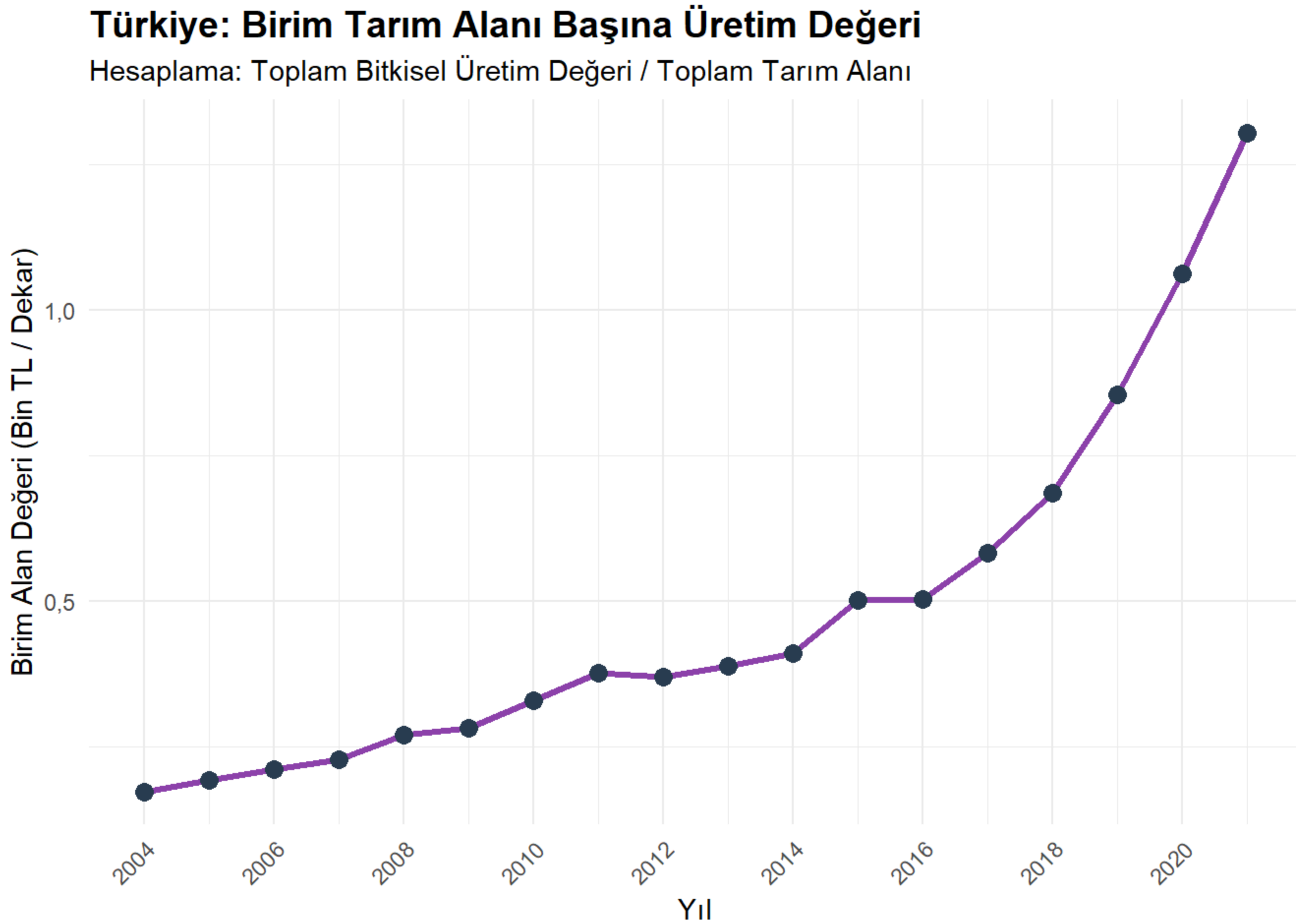

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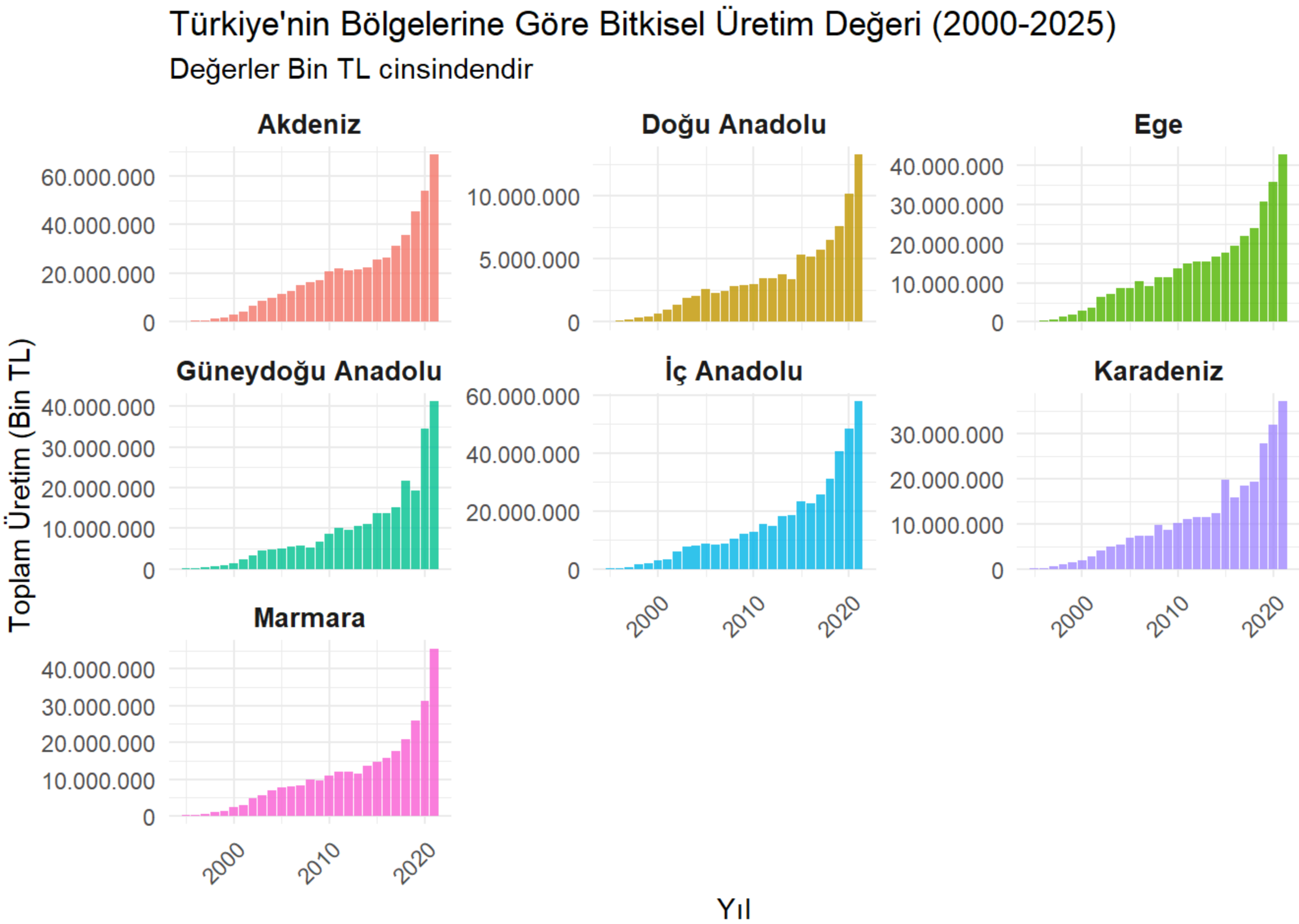

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